

Speaker 5 of 5 — Data Analysis 2: Competition, Skills & Scoring

Career Compass SG · Module 1 Assignment · Group 7

Your slot: 8:00 – 10:00 (2 minutes) · You close the presentation — and you do the live demo

Your job in one sentence

Deliver the payoff. Speaker 4 showed that volume and pay disagree; you show the two dimensions nobody else measures, then put all four together on screen and make the ranking change in front of the room. **The last thing the room hears is yours** — end on the message, not on a chart.

Where you sit in the flow

	Speaker	Topic
	1	Dashboard overview
	2	Technology stack
	3	Data cleaning
	4	Data analysis 1 — market & pay
→ YOU	5	Data analysis 2 — competition, skills & scoring

Speaker 4 hands you: "...the other two dimensions are where it gets interesting." **You close the presentation.** Nobody follows you.

Before you start

- The dashboard is already on the **Career Recommender** page — set up during Speaker 4's slot if you can do it without noise, otherwise click through during your first sentence
- Inputs pre-set: **3 years of experience, \$4,000 minimum salary**, all seniority levels
- Weight sliders at their defaults (30 / 30 / 25 / 15)

Script

Part A — Competition, the dimension nobody measures (8:00 – 8:35)

(Slide: the opportunity map — the four-quadrant bubble chart)

Every dot is a category. **Across is competition — applicants per seat. Up is volume.** Size is seats, colour is pay.

Competition varies 16-fold: Personal Care **0.7 applicants per seat**, F&B **1.0**, Social Services **11.3**.

Top left is where you want to be — volume without the queue. Top right is the trap. And this caveat is permanent: **these come from postings up to June 2023 only.**

Part B — Two findings that invert an assumption (8:35 – 9:05)

(Slide: hard-to-fill pay gap + the skill premium bar chart)

Two findings that reverse expectations.

Hard-to-fill means underpaid, not elite: reposted jobs pay **\$3,700 against \$3,850**. They're not elite roles — they're underpaid. That's your leverage.

And the skill premium is 4.6x — an AI title pays \$8,750, 127% above market; cleaning pays \$1,900, 51% below.

And keeping us honest: **IT postings fell 11.7% in six months** — the safest-looking track is cooling.

Part C — Live demo: putting the four together (9:05 – 9:45)

Four dimensions, four different units. We convert each to a **0-to-100 percentile rank** across 215 tracks, then weight them. That's the **Career Fit Score**.

I'm a **switcher with three years' experience who needs \$4,000 a month**.

(the page is already set — point at the result)

Top result: **IT at Mid level** — 69,000 postings, **\$6,500** median, four applicants per seat, **57% open to my three years**. These bars show why.

But watch. *(drag "Easy for me to enter" to maximum, drop "High pay")*

The entire top five changes — HR, Sales and Education at entry level. **Not one was in the previous list.** For someone who needs a job in six weeks, that's better advice.

The score is a set of sliders, not a verdict.

Part D — Close (9:45 – 10:00)

(Slide: the closing line)

The best-paying track and the easiest-to-enter track are almost never the same track. Every job board makes you guess which one you're looking at.

Now you can see both — and decide which one you need this year.

Thank you.

Numbers you must know cold

Figure	Value
Competition spread	0.7 to 11.3 applicants per seat — a factor of 16
Market-wide competition	3.77 applicants per seat
Hard-to-fill share / pay gap	4.1% of postings; \$3,700 vs \$3,850
Top skill premium	AI / Machine Learning, \$8,750, +127%
Bottom skill premium	Cleaning / Housekeeping, \$1,900, -51%
Premium ratio	4.6x
IT six-month growth	-11.7%
Fastest-growing categories	General Work +7%, Sciences / Lab +6%, Wholesale +5%
Steepest decline	Purchasing / Merchandising -23%
Tracks scored	215 , minimum 200 postings each
Default weights	demand 30 · pay 30 · low competition 25 · accessibility 15

The Career Fit Score, if you're asked to explain it

Component	Measure
Openings	percentile of $\log_{10}(\text{postings})$ — log, because 40,000 postings isn't twice as good a bet as 20,000
Pay	percentile of median advertised salary
Low competition	percentile of the <i>inverse</i> of applications ÷ vacancies, reliable window only
Accessibility	share of the track's postings open to your years of experience, re-ranked across surviving tracks

Weight sensitivity — the proof it isn't cosmetic

Rank	Balanced (default)	Money first	Easiest to enter
1	Information Technology – Mid	Information Technology – Senior	Human Resources – Entry
2	Information Technology – Senior	Information Technology – Mid	Sales / Retail – Junior
3	Sales / Retail – Junior	Information Technology – Management	Sales / Retail – Entry

Easiest to enter shares **no entries at all** with the default ranking.

Questions you own

"Aren't the weights arbitrary? You could get any answer you want."

That's the point, and it's why they're exposed rather than buried. There is no objectively correct weighting of money against speed-to-hire — that depends on the person. What we guarantee is that the four *inputs* are measured consistently and that you can see exactly how each one contributed.

"Why percentile ranks instead of the raw numbers?"

Because you can't add dollars to a ratio. Percentile ranks put all four on one 0-to-100 scale. The cost is that the score is relative — it tells you a track is better than others here, not that it's good in absolute terms.

"Your skill list is just keywords from job titles."

Correct, and it's labelled that way in the app. The dataset has no skills field, so we match titles against a 50-term dictionary spanning tech, business, trades, healthcare and services. It measures *what employers advertise for*, which is what a job seeker has to match — but it is not a skills census. A Python developer whose advert says "Software Engineer" isn't counted under Python.

"Isn't recommending F&B because it's uncontested bad advice?"

The tool never recommends on one dimension. F&B scores well on competition and volume and poorly on pay, and the user sees all four bars. If they weight pay highly, F&B drops out.

"Why a minimum of 200 postings per track?"

Below that the medians are too noisy to recommend anyone into. It's a judgement call, and it's in `config.py` as a named constant rather than buried in the code.

If the live demo fails: switch to the screenshot slides and narrate the same before/after ranking. **Do not debug.** The weight-sensitivity table above is your fallback — read it out.

Rehearsal checklist

- [] Timed at **2:00 or under**, including the slider drag
- [] You have dragged those sliders **on the presentation machine** at least three times
- [] You can name the four score components and why each is measured that way
- [] You can deliver the closing two sentences **without notes** — this is the last thing the room hears
- [] Screenshot fallback for the before/after ranking is on your slides
- [] You know who takes which question if the panel asks something outside your section